Spectro Scout

From process challenge
to a testable spectroscopy solution.

A materials-grounded decision-support copilot for early-stage in-line spectroscopy assessment.

EXPLAINABLE • LOCAL-FIRST • FEASIBILITY-FIRST

10-SLIDE PROJECT OVERVIEW

Prototype walkthrough · architecture · evaluation · limitations

The industrial measurement gap

Customers know the process pain—but not always the best optical method.

01 DELAYED FEEDBACK

Offline laboratory results can arrive after material has moved downstream.

02 COMPLEX MATRICES

Temperature, turbidity, scattering, fouling and formulation shifts complicate signals.

03 RISKY ASSUMPTIONS

A plausible technique is not proof of calibration transfer or hardware compatibility.

DESIGN OPPORTUNITY

Turn an unstructured customer question into a traceable first assessment: candidate methods, why they may fit, what is unknown, and how to test feasibility.

SUCCESS IS NOT “PICK A SENSOR”

Success is a defensible next experiment—and knowing when evidence is insufficient.

A structured path from intake to experiment

The app makes unknowns visible rather than silently filling them with assumptions.

1

Capture

Use case, target analyte, sample matrix, environment, measurement goal and reference method.

2

Assess

Rank candidate spectroscopy families using transparent local decision rules.

3

Challenge

Surface missing information, method limitations and compatibility gates.

4

Test

Propose a practical feasibility experiment with reference measurements and success criteria.

RESULTS WORKSPACE

- Ranked method candidates + heuristic fit scores
- Rationale, strengths, watch-outs and source references
- Priority clarifying questions
- Feasibility steps and proposed success criteria

Five spectroscopy families in scope

The right method depends on target chemistry, matrix, optical access and reference evidence.

NIR	**Composition / concentration trends**	Broad overlapping bands; calibration is matrix-specific.
MIR / FTIR	**Chemically informative fundamental bands**	Check water absorption, crystal, fiber and interface compatibility.
Raman	**Molecular fingerprinting**	Signal quality and fluorescence background can be decisive.
UV–Vis	**Defined absorption / optical-density signal**	Turbidity and optical density are not automatically target-specific.
Fluorescence	**Sensitive fluorescent species or marker**	Sensitivity alone does not establish selectivity or direct measurement.

Combination methods are hypotheses: establish single-method baselines before testing whether fusion adds value.

Lightweight, client-side architecture

A compact prototype with no external inference dependency.

USER INTERFACE

React 18 + React DOM

Intake form, sample cases, tabs and results dashboard.

DECISION ENGINE

src/engine.js

Deterministic rules, method library, questions and experiment plans.

DEVELOPMENT

Vite + Node.js + npm

Build, local development server and scenario regression checks.

RUNTIME FLOW

- Collect customer inputs
- Evaluate locally against explicit rules
- Render ranked candidates, unknowns and experiment plan

PRIVACY BY DESIGN

No API calls, no LLM, no database, and no upload of entered customer data in this archive.

Designed for explainability—not false certainty

The UI pairs an intake form with sample cases and an evidence-oriented results area.

APPLICATION INTAKE

- Use case description
- Target property / analyte + sample matrix
- Process environment and constraints
- Measurement goal / response expectations
- Reference method or laboratory data

ASSESSMENT OUTPUT

01 **Recommendations**
Ranked candidates, score, rationale and limitations

02 **Clarifying questions**
High-impact gaps with priority labels

03 **Feasibility plan**
Steps, reference comparison and success criteria

04 **Method guide**
Quick comparison of technique families

A completeness indicator nudges users to supply core facts before interpreting the output.

Six challenge cases encode important decision gates

Scenario checks verify the expected outcome shape—not scientific performance.

H01 **New API / powder blend**

Do not assume NIR calibration transfers; evaluate NIR and Raman separately.

H02 **Whey + opaque dairy**

Treat streams separately; compare interface geometries and matrix effects.

H03 **Fluorescence claims**

Sensitivity alone cannot prove direct glucose or viable-biomass measurement.

H04 **MIR + silica cable**

Do not approve MIR link until every optical component is compatible.

H05 **Unknown off-spec liquid**

Define target, reference labels and process constraints before selecting hardware.

H06 **Kidney study → surgery**

In-vitro research is not evidence of live clinical readiness.

Automated script: npm run test:scenarios • Assertions cover all H01–H06 cases and source/plan presence.

H01: calibration transfer is a testable risk

A new API and excipient blend changes the matrix; an old model cannot be presumed valid.

CANDIDATE PATH

1 NIR diffuse reflection

Evaluate the old calibration on new material without refitting; then build and validate a new model.

2 Raman as an independent comparison

EXPERIMENT DESIGN

- Collect representative batches across concentration + physical variation.
- Acquire spectra and independent API assays.
- Validate on held-out batches/runs.

DECISION GATES

Reference	Agree assay, uncertainty and acceptance limits.
Robustness	Represent particle size, density, flow and batch variation.
Combination	Test NIR + Raman only after single-method baselines.
Evidence	Retain combination only if held-out performance improves.

Evidence gates before real-world use

The prototype supports technical conversations; it does not certify an analyzer.

HEURISTIC SCORES

Ranking aids only—not probabilities, accuracy or detection limits.

CALIBRATION

Validate per formulation, matrix, instrument and operating envelope.

OPTICAL COMPATIBILITY

Confirm wavelength range, fiber, probe, connectors and interface materials.

REFERENCE QUALITY

Align spectra with independent assays; use held-out batches or runs.

HUMAN REVIEW

AFS experts must review safety, regulatory and installation constraints.

CLINICAL BOUNDARY

Research studies do not establish clinical readiness or surgical decision fitness.

Responsible-use principle: when evidence is insufficient, ask questions and design a test—not a confident guess.

A practical roadmap from prototype to pilot

Keep the core transparent; strengthen evidence, validation and operational readiness.

1 NOW

Run & review

Install dependencies; run build and H01–H06 scenario checks; review source references.

2 NEXT

Broaden testing

Add general-input tests, evidence schema and repeatable source/page citations.

3 THEN

Expert pilot

Define target-specific metrics, reference sampling, held-out validation and installation gates.

4 LATER

Operationalize

Only add backend, exports, CI or integrations when a clear requirement exists.

TAKEAWAY

Spectro Scout turns uncertainty into a transparent, evidence-led next experiment.